

# Supplementary Material for “GPU-Accelerated Uncertainty Quantification for Network Congestion: Coupled SDE Simulation and a Systematic Comparison of Rare-Event Estimators”

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## Scope

This supplement holds material referenced from, but not reproduced in, the main manuscript. Reviewer 1 asked us to compress the secondary modules; rather than delete these measurements we moved them here in full, so every number remains available and auditable. Nothing here is new evidence: each item supports a conclusion already stated and justified in the main text.

All measurements are on hardware we ran. No projected or modelled performance value appears in this supplement, consistent with the main manuscript.

## 1 Kernel Roofline Characterisation

### 1.0.1 Roofline Analysis

Figure 1 plots the roofline model for the SDE EM kernel on both Tesla T4 and A100 80 GB PCIe (FP32). Operational intensity is computed from the dominant operation—a dense  $(N_p \times n) \times (n \times n_{\text{local}})$  matrix-vector multiply per EM step—giving  $\text{OI} = (2N_p n^2 + 6N_p n) / (4(N_p n + n^2 + 2N_p n))$  FLOPs/byte, which ranges from 50–196 across the evaluated sizes.

The two devices have very different ridge OIs (T4: 25.4; A100: 161 FLOPs/byte), placing the same kernel in different regimes on each device (Table 1). On T4, whose lower ridge reflects its modest memory bandwidth (320 GB/s), all three sizes land *above* the ridge: the kernel is **compute-bound**, and efficiency climbs from 14% ( $n=500$ ) to 49% ( $n=5,000$ ). On A100, with its 1935 GB/s bandwidth and 312 TFLOPS FP32 peak, the ridge sits at 161 FLOPs/byte:  $n=500$  falls below it (memory-bound, 0.8%) and  $n=5,000$  reaches the transitional zone (5.0%). In both cases GPU utilisation grows monotonically with  $n$ , directly explaining why larger graph sizes benefit disproportionately from GPU execution and validating the workload-saturation story: at small  $n$  the SDE EM kernel is overhead- or bandwidth-limited; at  $n \gtrsim 5,000$  it approaches the compute roofline on T4 and enters the transitional zone on A100.

Table 1: Roofline characterisation of the SDE EM kernel on T4 and A100 (FP32, 512 paths, 100–200 steps). OI = FLOPs/Byte. Ridge OI: T4 = 25.4; A100 = 161.

| Device | $n$   | OI (FLOPs/B) | Achieved (GFLOPs/s) | Efficiency | Bound        |
|--------|-------|--------------|---------------------|------------|--------------|
| T4     | 500   | 63.2         | 1,137               | 14.0%      | compute      |
| T4     | 1,000 | 101.2        | 2,240               | 27.5%      | compute      |
| T4     | 5,000 | 196.0        | 3,967               | 48.7%      | compute      |
| A100   | 500   | 50.5         | 2,528               | 0.8%       | memory       |
| A100   | 1,000 | 84.2         | 8,567               | 2.7%       | transitional |
| A100   | 5,000 | 181.7        | 15,742              | 5.0%       | transitional |

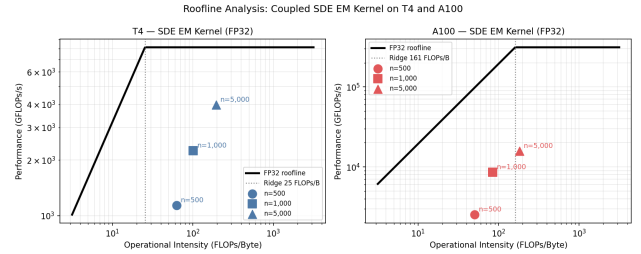


Figure 1: Roofline model for the SDE EM kernel on T4 (left, ridge 25 FLOPs/B) and A100 (right, ridge 161 FLOPs/B). Points move right and up with  $n$ , indicating growing compute intensity and efficiency. T4 kernels are compute-bound across all sizes; A100 kernels are memory-bound at  $n=500$  and transitional by  $n=5,000$ . Both devices confirm the same trend: GPU utilisation recovers as workload grows.

## 2 Single-GPU Kernel Phase Breakdown

Table 2 gives the `torch.profiler` phase decomposition of one fine-level path loop referenced from the main text, measured on a Tesla T4 over three seeds by device-side (CUPTI) kernel time. The degenerate single-rank all-reduce phase is omitted, as it measures nothing communicable and is not part of the single-GPU cost. The trend is that kernel launch/dispatch dominates at small graph size and the drift matrix multiply overtakes it as the kernels grow, while the Skorokhod reflection and the multi-level bookkeeping remain negligible throughout—so the reflected-boundary treatment and the telescoping sum,

Table 2: Single-GPU phase breakdown of one fine-level path loop (Tesla T4, three seeds, mean percentage of traced device-side kernel time). All values MEASURED. Raw per-seed data in `results/profiling_gpu/`.

| Phase                    | $n=500$ (%) | $n=5,000$ (%) |
|--------------------------|-------------|---------------|
| Kernel launch / dispatch | 65.3        | 50.5          |
| Drift matrix multiply    | 27.6        | 48.8          |
| Random-number generation | 5.0         | 0.4           |
| Skorokhod reflection     | 1.4         | 0.2           |
| MLMC level bookkeeping   | 0.7         | 0.1           |

Measured peak memory (same run):  $n=500$ , 13.6 MB ( $13.6\times$  the adjacency-only model);  $n=5,000$ , 109.8 MB ( $1.10\times$  the model).

Table 3: 4-GPU weak-scaling at large  $n$  on 4 $\times$ RTX 3090 PCIe ( $\varepsilon=0.05$ ). Single-GPU reference from A100 80 GB PCIe (Table ??); efficiency is a conservative bound due to device-generation difference. Efficiency recovers monotonically with  $n$ , confirming PCIe all-reduce amortisation at large workloads.

| $n$    | 1-GPU A100 (s) | 4-GPU 3090 (s) | Mem/GPU (MB) | Eff. (bound) |
|--------|----------------|----------------|--------------|--------------|
| 5,000  | 0.117          | 0.124          | 262          | 24%          |
| 10,000 | 0.296          | 0.212          | 973          | 35%          |
| 25,000 | 1.443          | 0.731          | 5,816        | 49%          |

the two problem-specific additions, cost almost nothing. The dominant device kernel at  $n=5,000$  is the drift matmul (`aten::mm` dispatching to `volta_sgemm`), at  $\approx 19$  ms per loop; every individual elementwise kernel—the reflection `clamp_min`, noise scaling, and level differencing—is under 0.25 ms.

A measured peak-memory pass in the same run shows the manuscript’s adjacency-only memory model ( $n^2$  bytes at single precision) is accurate only once the adjacency dominates: at  $n=500$  the true allocator peak is  $13.6\times$  the model (fixed ensemble and predictor–corrector buffers the model omits), but by  $n=5,000$  the ratio is  $1.10\times$ , confirming the model as a valid large- $n$  scaling driver rather than an absolute figure. Achieved SM occupancy is not reported: it requires Nsight Compute (ncu), which the CUPTI trace does not expose, and we do not estimate it.

### 3 Multi-GPU Measurements

The main text reports the multi-GPU contribution as *memory scalability* rather than strong-scaling speedup, and gives the headline weak- and strong-scaling figures. The per-configuration measurements behind those statements are collected here.

### 4 Halo-Exchange Staleness Sweep

The halo-exchange interval  $K$  trades communication volume against estimator bias, with the bias bounded by the

Table 4: Compute vs. Communication Breakdown on 4 $\times$ RTX 3090 (PCIe,  $n=1000$ , weak-scaling configuration). All-reduce time measured via NCCL event timing; compute time is total wall time minus all-reduce. The 43% efficiency at  $G=4$  is entirely communication-bound: compute time is flat across world sizes.

| GPUs | Compute (ms) | All-reduce (ms) | Total (ms) | Comm % | Efficiency |
|------|--------------|-----------------|------------|--------|------------|
| 1    | 75           | 0               | 75         | 0.0%   | 100%       |
| 2    | 75           | 1               | 76         | 1.3%   | 99%        |
| 4    | 65           | 110             | 175        | 62.9%  | 43%        |

Table 5: Blocking vs. overlapped NCCL all-reduce at  $G=2$  and  $G=4$  ( $n=2,000$ ,  $\varepsilon=0.05$ , 5 reps, PCIe hardware). Overlap Gain =  $(T_{\text{block}} - T_{\text{overlap}})/T_{\text{block}}$ ; Hidden fraction  $H = (T_{\text{block}} - T_{\text{overlap}})/T_{\text{allreduce}}$ . The larger all-reduce payload at  $G=4$  (more processes) increases the absolute gain.

| Metric                 | $G=2$ (RTX A5000) | $G=4$ (RTX 3090)  |
|------------------------|-------------------|-------------------|
| $T_{\text{block}}$     | $64.2 \pm 2.5$ ms | $91.3 \pm 2.9$ ms |
| $T_{\text{overlap}}$   | $57.4 \pm 0.3$ ms | $74.6 \pm 1.9$ ms |
| $T_{\text{allreduce}}$ | 11.5 ms           | 30.5 ms           |
| Overlap Gain           | 10.6%             | 18.3%             |
| Hidden fraction $H$    | 59.3%             | 54.9%             |
| Visible comm           | 4.7 ms            | 13.8 ms           |

Table 6: Halo-exchange interval sweep ( $G=4$  RTX 3090 PCIe,  $n=2,000$ ,  $\varepsilon=0.05$ , 5 seeds). Runtime and bias relative to  $K=1$  (fully correct baseline). Phase breakdown shows fraction of wall time per component; Sync  $\approx 0\%$  at all  $K$ .  $K=2$  is recommended: 22.5% runtime reduction at 2.9% bias.

| $K$ | Runtime (ms)   | RT Reduc. | Bias  | SDE%  | Book.% | Halo% |
|-----|----------------|-----------|-------|-------|--------|-------|
| 1   | $94.8 \pm 6.3$ | —         | —     | 51.1% | 7.2%   | 41.7% |
| 2   | $73.5 \pm 4.0$ | 22.5%     | 2.9%  | 68.5% | 8.0%   | 23.5% |
| 4   | $58.7 \pm 3.7$ | 38.1%     | 7.9%  | 77.9% | 9.3%   | 12.8% |
| 8   | $51.5 \pm 1.3$ | 45.7%     | 15.3% | 83.0% | 10.0%  | 7.0%  |

staleness term  $\mathcal{O}(L_{\text{lip}} K h_\ell)$  derived in the main text. The sweep below is the measurement supporting the recommended operating point  $K=2$ .

### 5 Data Availability

Every table in this supplement is reproduced by the scripts in the accompanying repository under `scripts/`, writing to `results/`. The per-seed raw data for the main manuscript’s evidence tables is archived under `results/baselines/`, `results/topology_matrix/`, `results/topology_matrix_stress/`, `results/validity_envelope/`, `results/rare_event_comparators/` and `results/significance/`.

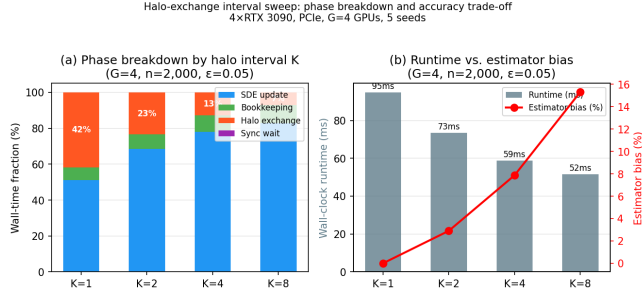


Figure 2: Halo-exchange interval sweep ( $G=4$  RTX 3090 PCIe,  $n=2,000$ ,  $\epsilon=0.05$ , 5 seeds). *Left*: stacked phase breakdown showing the fraction of wall time in each component (SDE update, MLMC bookkeeping, halo/all-reduce, sync wait) for  $K=1, 2, 4, 8$ . At  $K=1$  the halo phase accounts for 41.7% of total runtime;  $K=2$  halves this to 23.5%, proportionally increasing SDE compute time. *Right*: runtime (bars) and estimator bias (red line) as functions of  $K$ , showing the accuracy–efficiency trade-off.  $K=2$  offers 22.5% runtime reduction at only 2.9% bias.